

2023-Ph-H1-Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

An object accelerates from rest. Which graph represents its displacement over time?

Answer: A

Guidance for Students:

Starting from rest and accelerating means the object's speed increases over time. On a displacement–time graph, this shows as a **curved line with increasing steepness**.

Revision Tips:

- A straight line means constant speed.
- A curve getting steeper means increasing speed → acceleration.
- The gradient of an s - t graph gives velocity.
- From rest = initial gradient is 0.